

# Performance Analysis of 5G NR vRAN Platform and its Implications on Edge Computing

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## 2. BACKGROUND

Ultra-dense  
Bandwidth

Ultra-high  
Reliability

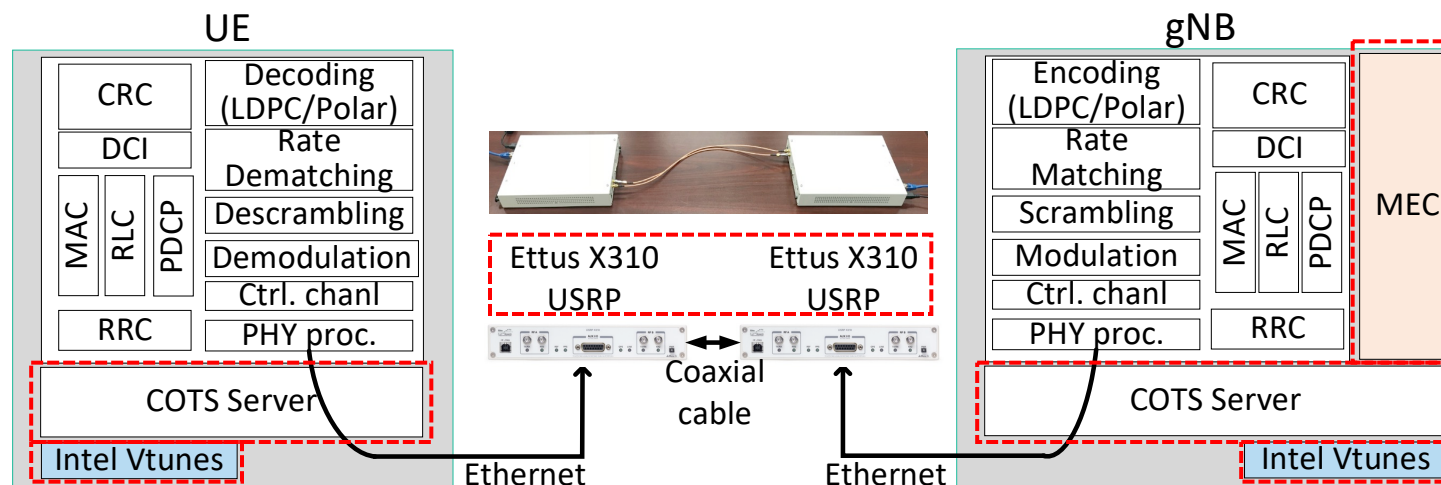


Ultra-low  
Latency

Network  
Function  
Virtualization

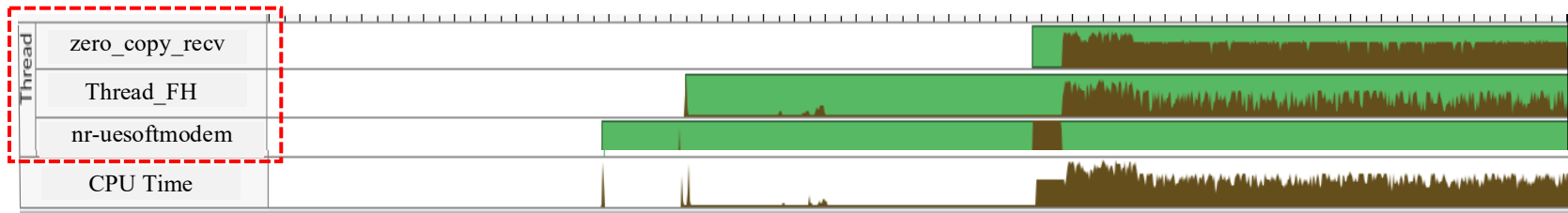
Mobile Edge  
Computing

### 3. TYPICAL RAN ARCHITECTURE



- We utilize the OpenAirInterface5g-nr platform for both 5G next generation NodeB (gNB) base station and User Equipment (UE). We choose Intel Xeon w-2194 for gNB and Intel Core i9-9900X for UE.
- We select PARSEC as our edge deployment to mimic edge-based video processing workloads. The Operating system used for both machines is Ubuntu 16.04 and both testbeds are implemented with a real RF front-end (Ettus X310 USRP). We use intel vtune to characterize the platform performance.

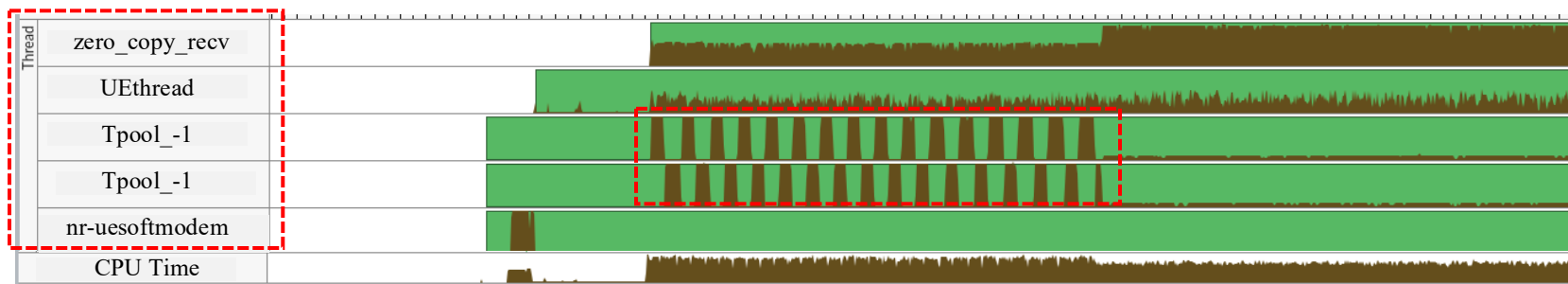
#### 4. CPU AND MEMORY USAGE OF gNB AND UE



CPU Time for 5G gNB threads - Downlink

- We observe that the dominant threads for OpenAirInterface5g-nr gNB are zero\_copy\_recv, thread\_FH and nr-softmodem.
- The zero\_copy\_recv thread is developed to mitigate the burden of the copy actions when transmitting data from Network Interface Card (NIC) to user space. It enables the direct transmission from NIC to user space instead of the two copy actions: from NIC to kernel space and from kernel space to user space.
- Thread\_FH does signal synchronization and detection while nr-uesoftmodem is used for the initialization.

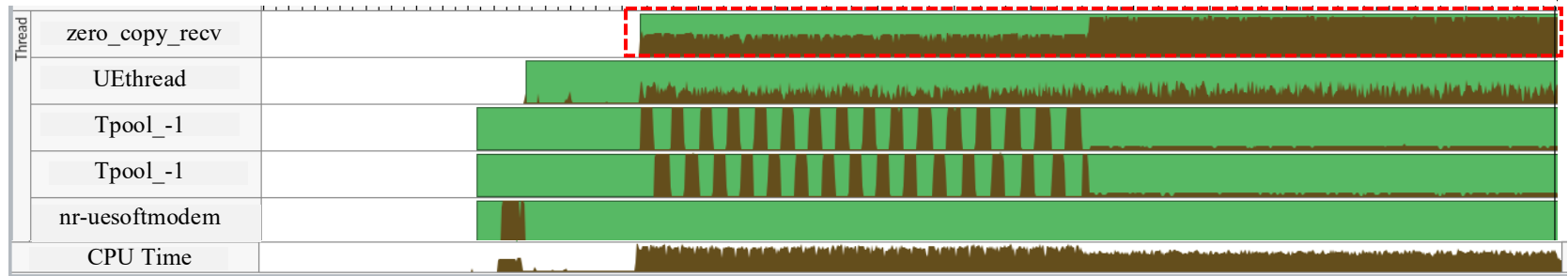
#### 4. CPU AND MEMORY USAGE OF gNB AND UE



CPU time for 5G UE threads - Uplink

- The dominant threads for OpenAirInterface5g-nr UE are zero\_copy\_recv, UThread, Tpool\_-1 and nr-softmodem.
- Tpool\_-1 which does signal synchronization and detection for OpenAirInterface5g-nr UE . This thread will occupy significant CPU. The optimization to reduce the CPU consumption of this thread is necessary for current nr system.

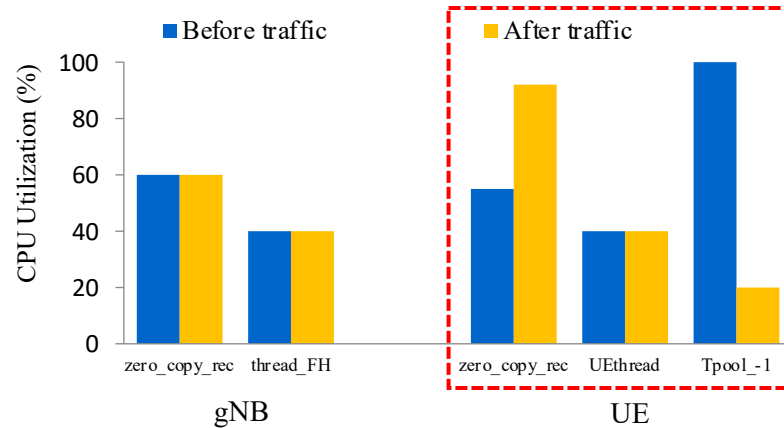
#### 4. CPU AND MEMORY USAGE OF gNB AND UE



CPU time for 5G UE threads - Uplink

- With the theoretical traffic burst for 5G, the traditional copy procedure will exhaust the device resource. However, we observe the CPU consumption of current zero copy reception is high whether there is traffic or not.
- To decrease the CPU consumption of the zero copy reception procedure would be a future direction for improving performance.

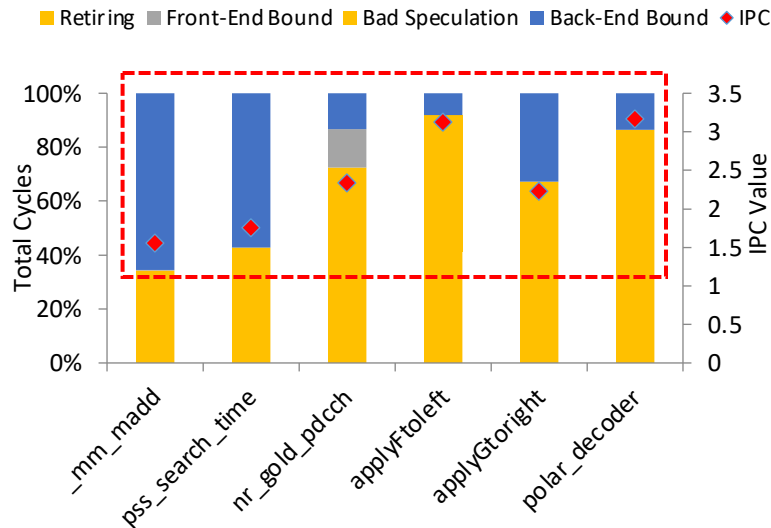
#### 4. CPU AND MEMORY USAGE OF gNB AND UE



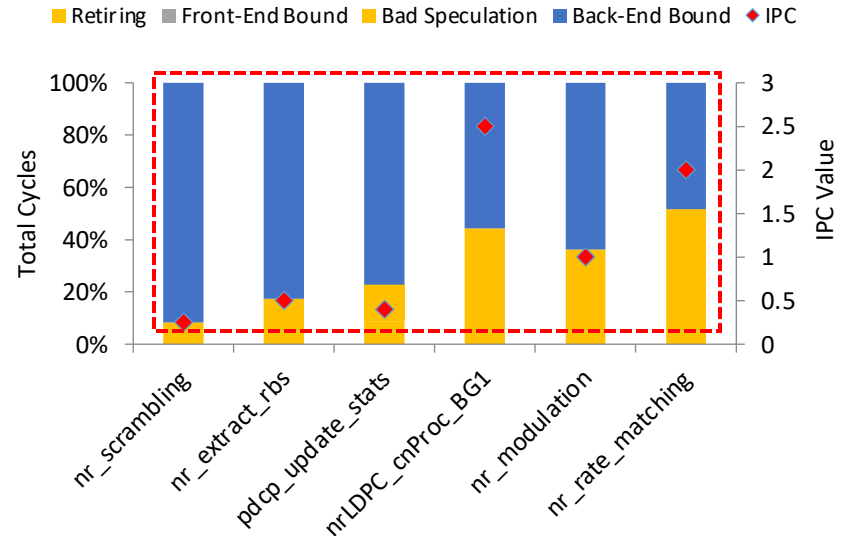
CPU usage before and after traffic income

- The UE's CPU utilization has an obvious change between the two stages. Before traffic coming in, the UE's CPU utilization is around 180%. After the traffic coming in, the UE's CPU utilization decreases to 160%.
- The reason for the decreasing is that the OpenAirInterface-nr still has some uncompleted submodules such as LDPC decoding and Polar decoding.

## 5. ARCHITECTURAL CHARACTERIZATION OF 5G FUNCTIONS



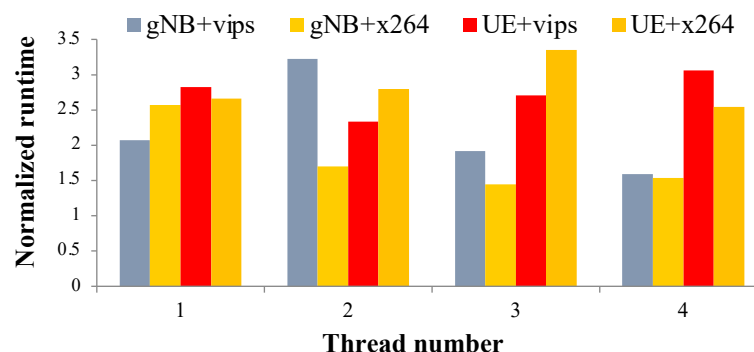
CPU time for 5G UE threads - Uplink



CPU time for 5G gNB threads - Downlink

- The main stall of OpenAirInterface5g-nr functions is concentrated on the Backend Bound. Backend Bound illustrates that no micro-ops are being delivered at the issue pipeline, due to lack of required resources in the Backend.
- The methods such as avoiding dependent arithmetic operations in a sequence; or better vectorization of OpenAirInterface5g-nr system may alleviate the severe Backend Bound performance limitation for OpenAirInterface5g system.

## 6. 5G NR AND MEC APPLICATIONS CO-LOCATION



### The MEC applications co-location runtime

- We observe that the performance of the OpenAirInterface5g-nr application does not degrade even though it is sharing the same core with the edge computing applications. The reason is that the threads created by OpenAirInterface-nr get the real time priority
- The edge computing applications require more time on completing the same-volume workload. The prolonged time for MEC applications range from 1.4X to 3.5X.
- When the edge computing applications have straight time-stamp will be a challenging case for the current co-run mechanism.